Device-Related Thrombus after LAA Occlusion in Case 3: Image Assessment and Haemodynamic Risk

Hugo Jal

Modelling of Organs and Systems, Universitat Pompeu Fabra

07/11/2026

Abstract

Is there a device-related thrombus (DRT)? No. In the CT, contrast-filled blood reaches the atrial face of the occluder and no thrombus can be seen on it. **Is the patient at risk of developing one?** Yes. A flow simulation of the left atrium (LA) with physiological flow and a mitral valve moving along its axis shows that the device sits in the quietest part of the atrium: the blood around it moves slowly, and the wall next to it has the lowest and most oscillating wall shear stress, which are the conditions linked to thrombus formation.

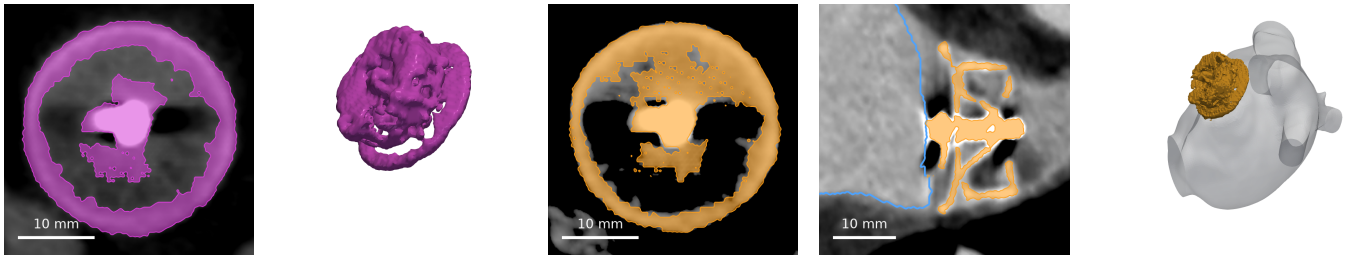


Figure 1: Segmentation. (a) U-Net device segmentation (magenta) on the plane of the device disc and (b) its 3D model. (c) Gamma-transformed CT on the same plane with the wire-frame segmentation (orange): rim, struts and hub. (d) CT along the device axis (LA on the left) with the LA (blue) and wire frame (orange): bright contrast blood reaches the atrial face of the device. (e) LA model (transparent) and device wire frame.

1 Introduction

Left atrial appendage occlusion (LAAO) closes the appendage with a device to prevent strokes in patients with atrial fibrillation. A known complication is a thrombus growing on the atrial face of the device. This device-related thrombus (DRT) is not frequent, but patients who develop it have a clearly higher risk of stroke [1]. In CT, a DRT appears as a dark (hypoattenuating) layer on the atrial side of the device [2]. Patient-specific flow simulations have linked DRT to slow blood and to low, oscillating wall shear stress around the device [3, 4]. In this report I answer two questions for Case 3: is there a DRT, and is the patient at risk of one?

2 Methods

2.1 Segmentation and mesh

The contrast CT (diastolic phase) was segmented in 3D Slicer (Fig. 1). The LA was obtained by manual thresholding of the contrast-filled blood, and then refined with manual painting and erasing, using the threshold range as a mask. The four pulmonary veins (PVs) were kept up to their first branch, the LA was closed at the mitral valve (MV), and the device was left out of the blood volume. A one-voxel opening removed small bridges, leaving a surface without holes. The device was segmented manually, and I tried two ways to extract only its wire frame. First, a pretrained promptable 3D U-Net (nnInteractive [6]), prompted with points on bright wires and on the blood around them. Alone it also filled the blood between the wires, so its output was intersected with the local wire contrast of the CT and corrected in 3D Slicer: it finds the rim and the hubs, but the struts merge into thick bands (Fig. 1a,b). Second, a gamma transform of the CT ($\gamma = 0.3$) and a filter for thin bright lines gave thinner struts and a closed rim (Fig. 1c). The surface was smoothed and repaired in MeshLab and Meshmixer, and each PV was cut perpendicular to its trunk and extended by 10 mm along it. The MV was cut on the annulus plane found in the CT, which was made the z axis, and the atrium was narrowed smoothly to a 5.3 cm^2 orifice (normal: 4–6 cm^2). The volume was meshed with tetrahedra in Gmsh (272 000 elements).

2.2 Flow simulation

The flow was solved in Ansys Fluent as transient and laminar, with blood as a Newtonian fluid ($\rho = 1060\,\mathrm{kg/m^3}$, $\mu = 3.5\,\mathrm{mPa\,s}$). The PVs received the course pressure curve. The MV received the unchanged course velocity curve, which on the 5.3 cm^2 orifice gives 80 mL per beat (5.5 L/min, a normal output [7]), and its longitudinal motion (up to 9 mm in systole), applied along the valve axis with a dynamic mesh (smoothing and remeshing) in which the outlet moves with the valve. With the time step of the boundary-condition files (0.01 s) I simulated four heartbeats and analysed only the last, free of start-up effects. All 350 time steps met the residual criteria (continuity $5 \cdot 10^{-3}$, velocity 10^{-3}), and the last two heartbeats gave the same flow volumes and almost the same wall maps.

2.3 Analysis

Results were exported at every time step and visualised in ParaView. From the wall shear stress $\boldsymbol{\tau}_w$ over one

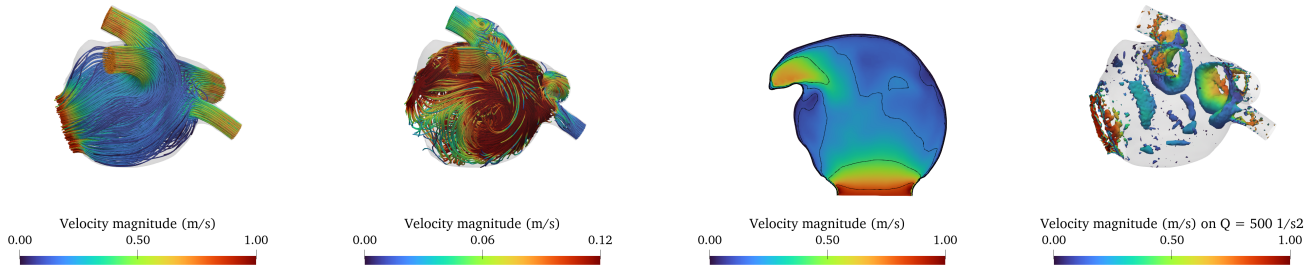


Figure 2: Flow in the last heartbeat. Streamlines from the pulmonary veins (a) at the peak of mitral flow and (b) in late diastole. (c) Velocity at the peak on a plane through the valve axis (valve at the bottom) and the device, which closes the bulge on the left wall. (d) Vortices at the peak ($Q = 500\,\mathrm{s}^{-2}$ [5]).

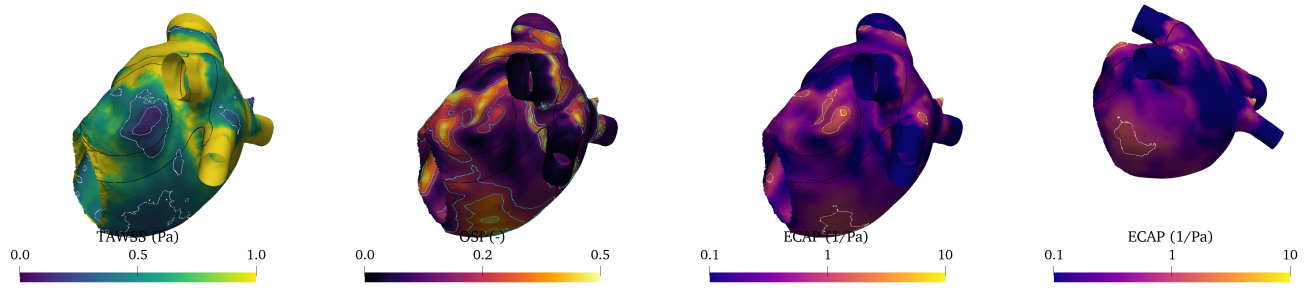


Figure 3: Wall shear metrics over the last heartbeat, seen from the device side (a–c) and from the front (d): (a) TAWSS, (b) OSI, (c, d) ECAP (log scale), each with isolines. Black lines: 10 and 20 mm from the device.

heartbeat of length T I computed

$$\mathrm{TAWSS} = \frac{1}{T}\int_0^T |\boldsymbol{\tau}_w|\,dt,$$

$$\mathrm{OSI} = \frac{1}{2}\left(1 - \frac{\left|\int_0^T \boldsymbol{\tau}_w\,dt\right|}{\int_0^T |\boldsymbol{\tau}_w|\,dt}\right), \quad \mathrm{ECAP} = \frac{\mathrm{OSI}}{\mathrm{TAWSS}}.$$

TAWSS is the average shear on the wall, OSI says how much the shear changes direction (0: never, 0.5: half of the time), and ECAP is high where the shear is both low and oscillating, which is the condition linked to thrombus formation [4]. To compare the device region with the rest of the atrium, I grouped the wall by distance to the device. For the flow I looked at streamlines, velocity on a plane through the device and vortex structures, shown with a Q-criterion of $500\,\mathrm{s}^{-2}$ as in left-atrium 4D flow MRI [5].

3 Results

3.1 Is there a thrombus?

No. In the CT slices the atrial face of the device is in direct contact with bright, contrast-filled blood (Fig. 1d).

3.2 Flow near the device

The flow is physiological: 80 mL per beat, inflow from all four veins (the small right inferior vein carries about 10 %) and velocities up to about 1 m/s in the mitral orifice. During rapid filling, the streams from the four PVs meet in the middle of the atrium and form a jet towards the MV, with vortex rings at the vein inlets (Fig. 2a,d). The blood at the entrance of the occluded appendage is the slowest of the section even at the peak of filling (Fig. 2c). In late diastole one large vortex fills the body, and the flow next to the appendage is again among the slowest (Fig. 2b).

3.3 Wall shear metrics

The wall maps (Fig. 3) show a region of low TAWSS and high OSI, and therefore high ECAP, at the entrance of the appendage where the device sits, inside the 10 mm line.

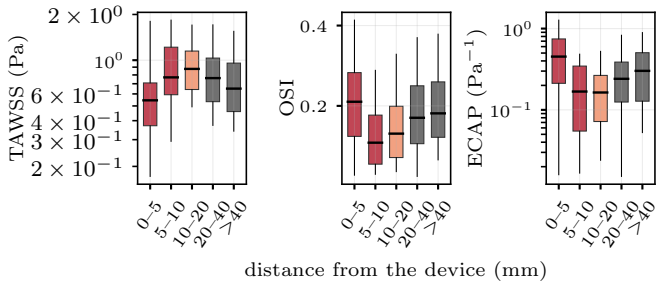


Figure 4: TAWSS, OSI and ECAP grouped by distance from the device (box: interquartile range, line: median, whiskers: 5th–95th percentile). Red: near the device.

Other, separate high-ECAP patches lie on the roof and on the lower wall, far from the device (Fig. 3d). Grouping the wall by distance (Fig. 4) shows the same trend: within 5 mm of the device the shear is the lowest and the most oscillating, and the median ECAP is about twice that of the rest of the atrium.

4 Discussion and conclusion

In this atrium, slow blood, low shear and oscillating shear all occur together on and around the atrial face of the device. Patient-specific simulations associate this pattern with a higher DRT risk [3], and in patients who did develop a DRT the thrombi appeared where these maps predicted [4].

Limitations: the boundary conditions are generic course curves, not measured for this patient, and the device is not in the flow domain as a solid object. So I compared regions of the same atrium, not fixed thresholds.

Conclusion. There is no device-related thrombus in the CT, but **the patient is at risk of developing one**: the device region has the slowest blood and the lowest, most oscillating shear of the atrium. Follow-up imaging is advisable.

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